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Adaptive Traffic Light Control System

BTCS 703-18

BACHELOR OF TECHNOLOGY
(Artificial Intelligence and Data Science)

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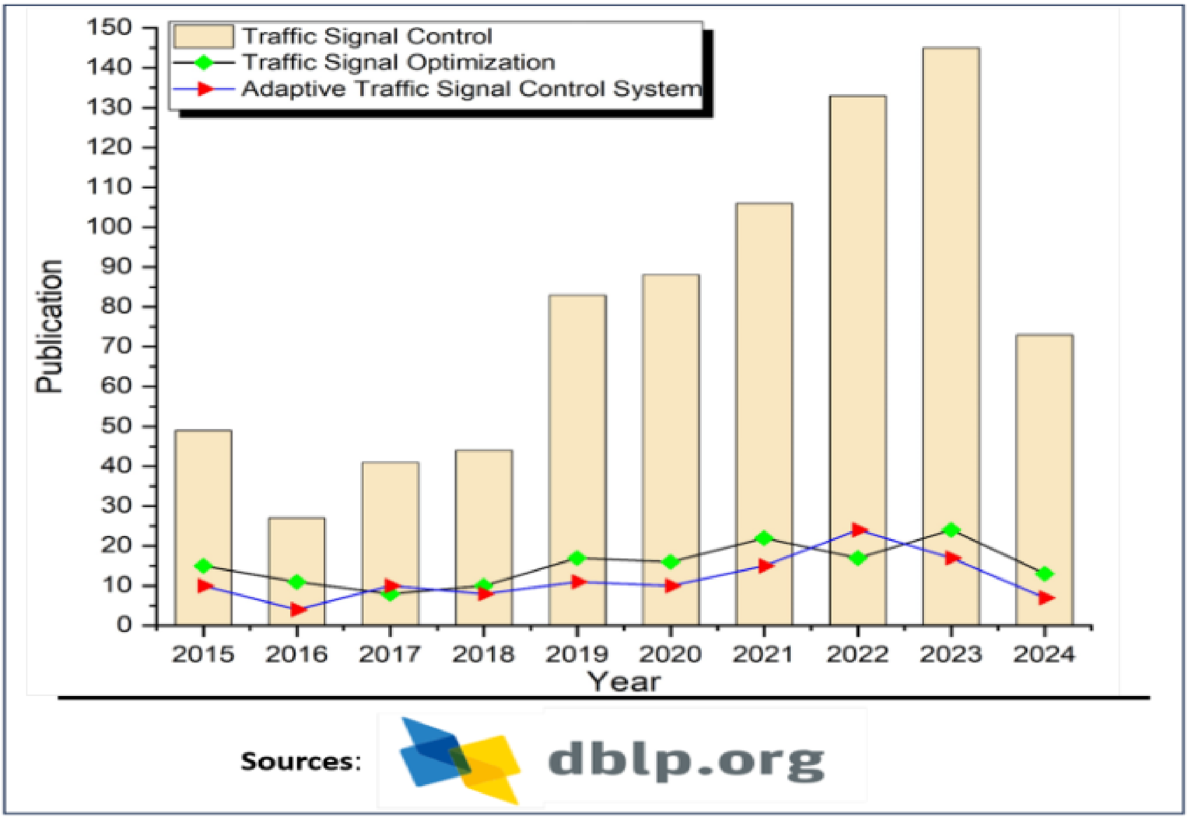
1.Introduction

1.1 Introduction

Controlling and managing traffic signals at to ensure vehicular traffic safety and a constant traffic flow is difficult for any dense urban area. The excessive number of vehicles transitioning from one place to another and the absence of proper flow management causes an interruption in the smooth traffic flow, resulting in lousy traffic congestion and inappropriate traffic control [1]. According to [2], fixed traffic signal (FTS) latencies account for about 10% of worldwide traffic delays. During the initial days of urbanization, many government agencies realized the need for traffic control at RIs and deployed old-age manual traffic signalized systems. In the manual Traffic Signal Control (TSC) system, the traffic officers modify the duration of the yellow, red, and green signals according to the traffic volume. This mode aims to achieve a smooth flow of traffic and safe passage for road commuters. Today, traffic light synchronization controls traffic at major RIs by providing the same amount of green signal time to all directions during the one cycle known as FTS. Various papers [3,4] suggested that developed countries suffer over 295 million traffic hours of delay with Fixed Traffic System. Thus, detailed estimations of time-dependent delays are needed for better traffic control and management.

Congestion results from the complex traffic system architecture's inability to coordinate or correlate the timing of traffic lights with the traffic volume, which is dynamic with the season and time of day. Thus, detailed estimations of time-dependent delays are needed for better traffic control. Congestion results from the complex traffic system architecture's inability to coordinate or correlate the timing of traffic lights with the traffic volume, which is dynamic with the season and time of day. The graph shows publication trends on Traffic Signal Control, Traffic Signal Optimization, and Adaptive Traffic Signal Control Systems from 2015–2024. Overall traffic signal control research increased sharply after 2018, peaking in 2022–2023, while optimization and adaptive systems had fewer but gradually rising publications with small peaks in recent years.

Figure 1. Publication status of Traffic Signal Control with its application as an optimization and Adaptive Traffic Signal Control system in the last decade.



1.2 Main Idea

The main idea of our ATLCS consists in synchronizing a number of traffic lights controlling consecutive junctions by creating a delay between the times at which each of them switches to green in a given direction. Such a delay is dynamically updated based on the number of vehicles waiting at each junction, thereby allowing vehicles leaving the city centre to travel a long distance without stopping (i.e., minimizing the number of occurrences of the ‘stop and go’ phenomenon), which in turn reduces their travel time as well. The performance evaluation of our ATLCS has shown that the average travel time of vehicles traveling in the synchronized direction has been significantly reduced (by up to 39%) compared to non- synchronized fixed time Traffic Light Control Systems. Moreover, the overall achieved improvement across the simulated road network was 17% [5]. The concept of the Smart City has emerged in recent years as a futuristic vision of cities building sustainable ecosystems, while promoting citizen welfare and economic growth. A Smart City fosters the use of advanced ubiquitous information communication technologies (ICTs) to smartly and efficiently monitor and manage its critical assets such as energy, water, and transportation infrastructure. Such an ambition can become a reality only with joint efforts from governmental, industrial, academic and social actors. Building smart cities, however, strongly depends on various enabling advanced technologies (e.g., sensors, Internet of Things (IoT), 5G networks, the cloud, Artificial Intelligence, connected vehicles, etc.) to support sustainable developments such as smart buildings and energy, and smart and green transportation.

2. Brief Literature survey

2.1 Traffic Data Collection in Urban Cities

Many researchers have worked on traffic management to reduce congestion and delays in urban areas. Traffic congestion has become one of the biggest challenges in almost all major Indian cities. With rapid urbanization, increasing population, and rising vehicle ownership, city roads are unable to cope with the growing demand. Daily commuting has become not only time-consuming but also stressful, leading to loss of productivity, increased fuel consumption, and higher levels of pollution. Several national and international surveys, including reports from [6] reveal the severity of this issue. They provide valuable insights into how long it takes to travel short distances during peak hours, the number of hours wasted annually in traffic, and the percentage of congestion compared to free-flowing conditions. These statistics give a clear picture of how much urban life and economic activity are affected by inefficient traffic flow. The table below summarizes key facts and figures for some of India's most congested cities.

City	Average Time to Travel 10 km (Peak Hours)	Hours Lost Annually in Traffic / Congestion	Congestion Level / % (how much extra time compared to free-flow)
Kolkata	~ 34 min 33 sec	110 annual hours lost	Congestion quite high (in worst 2nd slowest city in India/world)
Bengaluru	~ 34 min 10 sec	~ 117 hours lost annually	Among top congested globally
Pune	~ 33 min 22 sec	~ 108 hours lost annually	High congestion
Chennai	~ 30 min 20 sec	~ 94 hours lost annually	Moderate-high congestion
Mumbai	~ 29 min 26 sec to cover 10 km	~ 103 hours lost annually	Congestion level ~ 35–54% depending on area/time
Delhi	~ 21 min 40 sec (for 10 km) in city center average	~ 191 hours driving during peak hours; number lost due to congestion less.	Traffic slow; speed ~ 24 km/h during rush hour in city center.

2.2 Conventional Traffic Control System

Conventional traffic control systems have been based on either fixed-time or simple adaptive algorithms using proximity sensors. In both cases, the traffic signal sequence is predetermined with small variations based on the time of the day or binary detection of vehicles using proximity sensors. [7] These conventional fixed traffic light controllers have limitations and are less efficient because they use a hardware, which functions according to the program that lacks the flexibility of modification and adaptation on a real time basis. Thus due to the fixed time intervals of green and red signals there is excess and unnecessary waiting time on roads and vehicles consume more fuel. This eventually adds up to the environmental pollution and creates several health issues among the people on road and residing nearby. Also these conventional traffic light control systems do not have any provisions to provide any information on traffic densities on various roads, which leads to traffic congestions, and traffic congestion has a significant impact on network-wide fuel consumption by causing frequent stops of vehicles and extending their travel times in the traffic network. Due to this fact, efficient or optimal traffic signal control is gaining increased attention as an effective means to reduce network-wide fuel consumption and alleviate traffic congestion within the traffic network.

2.3 Adaptive Traffic Control System

In order to further improve the network-wide energy efficiency of the vehicles, a new traffic signal control method is proposed in this paper. By utilizing real-time traffic data from advanced sensors, such as roadside cameras and radars, it is possible to obtain rich and diverse traffic information, including the number of vehicles, total vehicle speed, and total vehicle waiting time in each traffic phase. The proposed novel control method employs various combinations of this diverse traffic information to control traffic signals at multiple intersections optimally. Simulation results based on a real-world traffic network with actual traffic data reveal significant improvements in the network-wide energy efficiency compared to a conventional fixed-time method and a sensor-based actuated coordinated control method. It is also shown that incorporating additional information on vehicle type further improves network-wide fuel efficiency. Optimal control of traffic signals at multiple intersections using diverse traffic information is novel and will pave the way for similar research as more efficient and cost-effective advanced sensors are developed. Few real world examples of the use of Adaptive Traffic Control System are given in the following table.

Location	System / Project	Reported Results
Bengaluru, India	Bengaluru Adaptive Traffic Control System (BATCS) – launched in 2024 at 60 junctions (planned for 165). AI-based adaptive signals at key corridors.	Travel time reduced by up to 33%, throughput increased 18-30%, average vehicle speeds improved.[8]
London, UK	TfL's Real Time Optimiser (RTO) / FUSION system covering ~6,400 junctions & crossings. Uses real-time traffic & multimodal data.	Improved journey times, more responsive to incidents, smoother traffic for buses, cars, cyclists, and pedestrians.[9]
Los Angeles, USA	ATSAC (Automated Traffic Surveillance and Control) – one of the largest adaptive systems, controlling 4,500+ signals.	Reported 12% reduction in travel time, fuel savings, reduced congestion along major corridors.[10]
Pittsburgh, USA	Surtrac (Scalable Urban Traffic Control) – AI-based decentralized ATCS deployed in East Liberty	Reduced travel times by 25%**,** wait times by 40%**,** emissions cut by 21%**.**[11]

3. Problem Formulation

Traffic congestion is a long-standing issue that urban planners and engineers have attempted to mitigate using various technologies. Urban traffic congestion has become a critical issue in cities worldwide, especially in India, where rapid urbanisation and an ever-increasing number of vehicles have overwhelmed existing road infrastructure. Traditional traffic signal systems typically operate on fixed-time schedules, which are unable to adapt to real-time traffic conditions. As a result, vehicles often experience unnecessary waiting times at intersections, leading to longer commute durations, increased fuel consumption, higher vehicular emissions, and greater frustration among commuters. Semi-actuated or manually controlled traffic signals offer only limited relief, as they cannot efficiently optimise traffic flow across multiple intersections simultaneously.

According to the TomTom Traffic Index 2024**,** Indian cities such as Bengaluru, Mumbai, and Kolkata are among the most congested globally, with average commute times increasing by 30–35% during peak hours. This highlights the urgent need for smarter traffic management solutions capable of responding dynamically to changing traffic conditions. Fixed-time traffic systems are not only inefficient in reducing congestion but also fail to prioritise emergency vehicles or adapt to special events, accidents, or weather-related disruptions.

Therefore, our core problem is to design an Adaptive Traffic Light Control System that can continuously monitor real-time traffic conditions, dynamically adjust signal timings, and reduce congestion, waiting times, and environmental impact at urban intersections.

4. Objectives

The primary objective of this project is to design and implement an Adaptive Traffic Control System (ATCS) that optimises traffic signal operations based on real-time traffic conditions, thereby reducing congestion, waiting time, and environmental impact at urban intersections. Unlike traditional fixed-time or semi-actuated traffic signals, the proposed system aims to respond dynamically to fluctuating traffic patterns, ensuring smoother traffic flow and more efficient use of road infrastructure.

The specific objectives of this project include:

1. **Real-time Traffic Monitoring:** To gather and process traffic data from sensors, cameras, or publicly available datasets, including vehicle density, queue lengths, and flow patterns at multiple intersections. To implement algorithms that continuously monitor traffic conditions and detect peak and off-peak periods.
2. **Dynamic Signal Timing:** To develop a system that can adjust green, yellow, and red signal durations dynamically based on real-time traffic conditions. To optimise signal sequences for multiple intersections in coordination, preventing unnecessary waiting times and bottlenecks.
3. **Reduction of Travel Time and Congestion:** To minimise vehicle waiting times at intersections by applying adaptive algorithms that respond to changing traffic loads.
4. **Environmental and Economic Benefits:** To reduce vehicular emissions by decreasing idle time at red lights, contributing to cleaner urban air, lower fuel consumption and operating costs for commuters and transport operators.
5. **Future Scalability and Smart City Integration:** To ensure that the system is scalable and can be integrated with IoT devices, GPS-based vehicle tracking, provide a foundation for further enhancements such as emergency vehicle prioritisation, pedestrian-friendly adaptive signals, and predictive traffic modelling.

5. Methodology/ Planning of work

According to [], The proposed system aims to develop an intelligent traffic monitoring and management system using deep learning and IoT-based approaches. The methodology involves a combination of real-time object detection, data analytics, and AI-driven decision-making to enhance traffic flow, improve safety, and reduce congestion in urban environments. The system architecture integrates computer vision, edge computing, and cloud-based analytics for efficient data processing.

1. **Data Collection and Preprocessing:**
   IoT-enabled sensors, cameras, and GPS trackers are deployed at intersections to collect traffic data. Data attributes include vehicle count, speed, congestion levels, and weather conditions. The raw data undergoes preprocessing techniques such as cleaning, normalization, and feature selection for model training.
2. **Machine Learning-Based Traffic Prediction:**
   Supervised learning models (Random Forest, Support Vector Machines, and Deep Learning techniques like CNNs and LSTMs) are trained on historical and real-time traffic data. Traffic signal timings dynamically adjust based on real-time congestion data Anomaly detection is implemented for incident detection and emergency response.
3. **IoT and Cloud-Based Integration:**
   Cloud computing is used for real-time traffic data processing and model updates. IoT-enabled traffic control units send live data to cloud-based servers. Edge computing is integrated for low-latency response in critical intersections.
4. **Performance Evaluation Metrics:**
   Key Performance Indicators (KPIs) such as average waiting time, vehicle throughput, congestion index, and accident response time are analyzed. Comparative analysis is performed against traditional fixed-time traffic light systems. Real-world traffic simulation using SUMO (Simulation of Urban Mobility) validates the system’s effectiveness.

Facilities required for proposed work

The system architecture of the proposed smart traffic control system consists of multiple interconnected layers, ensuring seamless data flow, processing, and decision-making.

System Components:

1. **Traffic Data Acquisition Layer:**
   IoT sensors, CCTV cameras, and GPS trackers collect live traffic data. Vehicle detection, lane occupancy, and speed monitoring are performed.
2. **Data Processing and Storage Layer:**
   The collected data is transmitted to cloud servers for preprocessing. Data cleansing, feature extraction, and predictive model updates occur in this stage.
3. **Traffic Prediction and Decision-Making Layer:**
   Machine learning models predict congestion levels and traffic density. Adaptive traffic light control algorithms adjust signals dynamically.
4. **Communication and Actuation Layer:**
   The processed data is sent to traffic signal controllers for real-time adjustments. Alerts are generated for emergency vehicles, congestion zones, and accident-prone areas.
5. **User Interface and Monitoring Layer:**
   Traffic authorities monitor the system via a web or mobile dashboard. Citizens receive live traffic updates through mobile applications.

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